

Database Design Document

CodeGrade: Smart Student Assignment & Code
Reviewer

Group - 11

Batch - 8B

1. Overview:-

The CodeGrade database is designed using a relational model and will be implemented via Entity Framework Core (Code-First approach) on SQL Server. The database context will be strictly managed by a backend Web API to ensure data security. It will utilize Foreign Key constraints with Cascade Delete enabled for required relationships to maintain strict data integrity.

2. Entity Relationship Schema:-

Table: Users

Will store authentication and profile data for all platform users. Role-based access will dictate platform permissions.

Column Name	Data Type	Constraints	Description
Id	int	Primary Key, Identity	Unique identifier for the user.
Name	nvarchar(max)	Required	Full name of the user.
Email	nvarchar(max)	Required	User's email address for login.
Password	nvarchar(max)	Required	Hashed password string.
Role	nvarchar(max)	Required	Enum/String: 'Teacher' or 'Student'.

Table: Classrooms

Will represent a coding class created and managed by a Teacher.

Column Name	Data Type	Constraints	Description
Id	int	Primary Key, Identity	Unique identifier for the class.
Name	nvarchar(max)	Required	Title of the classroom.
Description	nvarchar(max)	Optional	Overview of the class subject.
JoinCode	nvarchar(max)	Required, Unique	Code used by students to enroll.
TeacherId	int	Foreign Key	References Users(Id). Cascade Delete.

Table: Enrollments

Will establish a Many-to-Many relationship between Students and Classrooms.

Column Name	Data Type	Constraints	Description
Id	int	Primary Key, Identity	Unique enrollment record ID.
StudentId	int	Foreign Key	References Users(Id). Cascade Delete.
ClassroomId	int	Foreign Key	References Classrooms(Id). Cascade Delete.
JoinedDate	datetime2	Required	UTC timestamp of enrollment (Converted to local IST on the client frontend).

Table: Assignments

Will store tasks created by teachers within a specific classroom.

Column Name	Data Type	Constraints	Description
Id	int	Primary Key, Identity	Unique assignment identifier.
Title	nvarchar(max)	Required	Name of the assignment.
Description	nvarchar(max)	Required	Instructions for the assignment.
AllowedLanguage	nvarchar(max)	Required	e.g., 'Any', 'python', 'csharp'.
ClassroomId	int	Foreign Key	References Classrooms(Id). Cascade Delete.

Table: Submissions

Will contain the actual code submitted by students. The grading columns are designed to be nullable so they can be populated asynchronously by the automated AI webhook after the initial submission.

Column Name	Data Type	Constraints	Description
Id	int	Primary Key, Identity	Unique submission identifier.
CodeText	nvarchar(max)	Required	The raw code written by the student.
Language	nvarchar(max)	Required	Language the code was written in.
SubmissionDate	datetime2	Required	UTC timestamp of submission (Converted to local IST on the client frontend).
StudentId	int	Foreign Key	References Users(Id). Cascade Delete.
AssignmentId	int	Foreign Key	References Assignments(Id). Cascade Delete.
Grade	int	Nullable	Reserved to be populated by the AI evaluation out of 100.
AiFeedback	nvarchar(max)	Nullable	Reserved for AI's detailed critique.